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12.4 Classwork: de Moivre, Trigonometric Identities and Roots of Unity (no calculator)

Answer all three problems in the spaces provided. Show your working. Give all answers as exact values (surds and fractions of π), not decimals.

1. Let θ be any real number.

- (a) Use de Moivre's theorem to write $(\cos \theta + i \sin \theta)^3$ in terms of $\cos 3\theta$ and $\sin 3\theta$.
- (b) Expand $(\cos \theta + i \sin \theta)^3$ using the binomial theorem, simplifying the powers of i .
- (c) By equating real parts, show that

$$\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta.$$

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2. (a) Using your expansion from Problem 1, equate imaginary parts to find an expression for $\sin 3\theta$ in terms of $\sin \theta$ only.

- (b) Hence solve the equation

$$8 \cos^3 \theta - 6 \cos \theta = 1, \quad 0 \leq \theta < \pi,$$

giving all solutions as exact multiples of π .

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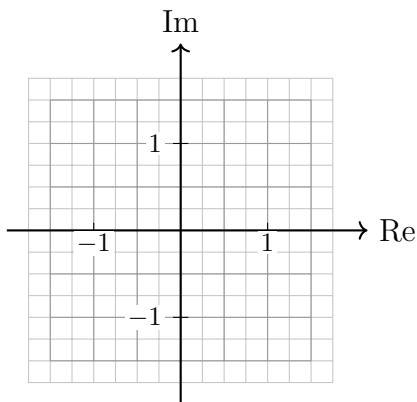
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3. Let ω be a complex number satisfying $\omega^5 = 1$.

- (a) Find all five solutions of $z^5 = 1$, giving each in the form $e^{i\theta}$ with $-\pi < \theta \leq \pi$.
- (b) Show that the five solutions sum to zero.
- (c) The five solutions are plotted on an Argand diagram. Describe the figure they form, stating its centre and the distance of each vertex from the origin.



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